

AdventureWorks – SQL Assignment #2

Business Brief Edition · 60 tasks framed as real stakeholder requests. Read the request, choose the technique, and write the query.

Restriction: Do not use window functions (ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD, etc.).

Section A – Joins & Filtering (1–10)

- Q1. The US sales team needs a contact list. Pull every US-based customer's full name, email, city, and country.
- Q2. Merchandising wants the full assortment. List every product with its subcategory and category.
- Q3. The e-commerce team needs an order-level view. Compile every internet sales order with its customer and product.
- Q4. Marketing is curating a themed range. Find all products whose English name contains the word 'Mountain'.
- Q5. Leadership is reviewing our three core markets. List all customers in the United States, Canada, or Australia.
- Q6. Finance is closing the 2013 books. Show all products that recorded sales during 2013 (1 Jan – 31 Dec).
- Q7. The digital team is auditing online gaps. Find every product that has never generated an internet sale.
- Q8. Support is segmenting alphabetically. Provide all customers whose last name begins with 'B'.
- Q9. The CRM team wants cleaner data. Flag every customer with no email address on file.
- Q10. Pricing is reviewing the mid-range. Display all products with a list price between \$500 and \$1,500.

Section B – Aggregations & Grouping (11–20)

- Q11. The board wants the revenue trend. Calculate total sales for each year.
- Q12. Account management is gauging engagement. Report the total number of orders placed by each customer.
- Q13. Leadership wants to recognise top accounts. Identify the ten customers with the highest total sales.
- Q14. Pricing wants a benchmark. Calculate the average unit price for each product category.
- Q15. Regional planning needs the footprint. Count the number of customers in each country.
- Q16. For the strategy review, identify the flagship categories – those generating more than \$1,000,000 in sales.
- Q17. Territory managers need benchmarks. For each territory, report the highest, lowest, and average order amount.
- Q18. Identify high-density markets – countries where we serve more than 100 customers.
- Q19. Supply chain needs demand volumes. Calculate the total quantity sold for each product.

Q20. Merchandising is reviewing breadth. Identify categories that carry more than 50 products.

Section C – Built-in Functions & CASE (21–25)

Q21. Standardise records for a mail merge. Present each customer's full name in uppercase and email in lowercase.

Q22. The insights team is building profiles. Calculate each customer's current age from their date of birth.

Q23. Finance wants monthly trends. Extract year and month from order date and total sales by year-month.

Q24. Marketing wants price tiers. Classify each product as Premium, Standard, or Budget based on list price.

Q25. Build a value segmentation. Label each customer High, Medium, or Low Value based on total spend.

Section D – Subqueries (26–35)

Q26. Pricing is studying the premium end. Find products whose list price exceeds the catalogue-wide average.

Q27. Find above-average customers – those whose total spend beats the average customer spend.

Q28. Leadership wants the headline winner. Display the product(s) with the highest total sales amount.

Q29. Find the most active buyers – customers who placed more orders than the average per customer.

Q30. Spotlight strong regions. Show territories whose total sales exceed the average across all territories.

Q31. Rank the portfolio. Find products whose total sales are higher than the average product sales.

Q32. For a turnaround review, find territories whose total sales fall below the overall average.

Q33. Find big-ticket buyers – customers whose largest single order exceeds the company's average order amount.

Q34. Find the widest-reach products – those sold to more distinct customers than the average product.

Q35. Find premium markets – countries where average customer spend is above the company-wide average.

Section E – Correlated Subqueries & EXISTS (36–45)

Q36. Within each category, find the standouts – products whose total sales beat their own category's average.

Q37. We're planning a bike push. Find customers who have never purchased a bike.

Q38. Find each market's leaders – customers whose total purchases exceed the average spend of their country.

Q39. Find globally popular products – those sold to customers from at least 5 different countries.

- Q40. The Australia team wants gaps. Find products that have never been sold in Australia.
- Q41. Spot regions with no marquee account. Find countries where no customer has generated more than \$10,000 in sales.
- Q42. Find cross-sell targets. Find customers who have purchased from only one product category.
- Q43. Find perennial sellers — products that have sold in every year present in the dataset.
- Q44. Find the most engaged customers — those who have purchased from every product category.
- Q45. Find fully-activated categories — those in which every product has at least one sale.

Section F — CTE-Based Reporting (46–50)

- Q46. Using a CTE, build a high-value list — customers with total sales above \$10,000.
- Q47. Using a CTE, find cornerstone markets — countries each contributing more than 10% of company revenue.
- Q48. Using multiple CTEs, compare yearly revenue and flag any year that declined versus the prior year.
- Q49. Using a CTE, compute lifetime revenue and assign a tier: Platinum >20,000, Gold 10,000–20,000, Silver 5,000–10,000, Bronze <5,000.
- Q50. Using CTEs and joins only, produce a customer summary: name, country, # orders, total sales, average order value, distinct products, distinct categories.

Challenge Round — Interview Level (51–60)

- Q51. Find cross-category champions — customers who purchased from more than 3 different categories.
- Q52. Find any customer whose total spend exceeds the combined total sales of all Canadian customers.
- Q53. Find truly universal products — those sold in every country where AdventureWorks has customers.
- Q54. Find year-after-year loyalists — customers who purchased in every year present in the dataset.
- Q55. Find category completists — customers who purchased every product within a category (e.g., Mountain Bikes).
- Q56. Find underperforming categories — those where more than 50% of products have never been sold.
- Q57. Find category revenue drivers — products contributing more than 5% of their category's revenue.
- Q58. Find the highest-revenue category — without using TOP.
- Q59. Find strategically critical accounts — customers whose spend is more than 1% of total company sales.
- Q60. Produce a complete executive sales report: Year, Country, Total Customers, Total Orders, Total Sales, Average Order Value, Most Sold Category, Highest Revenue Category — using only joins, aggregations, subqueries, CTEs, and built-in functions.

Submission Rules

1. Use meaningful, business-readable aliases.
2. Do not use SELECT * – name the columns you need.
3. Format SQL cleanly and consistently.
4. Add a short comment above each query.
5. Use CTEs where required.
6. Do not use window functions.
7. Submit one .sql file containing all 60 solutions.
8. Include result screenshots for at least 15 queries.